

Boundary Conditions and Open or Closed Endpoints

- Exactly one rule may own a boundary value. The rule whose condition carries $\leq$ or $\geq$ is the one that owns it; the other uses $<$ or $>$ and stops just short.
 - If two conditions both claimed the same input, the definition would hand back two different outputs for one input — and that is not a function at all.
 - After solving an equation from one branch, **test each answer against that branch's condition**. A value can satisfy the formula and still not be a solution.
 - Where a student's work is shown, it contains exactly one error — and the arithmetic is usually fine.
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1. A piecewise function is defined by

$$f(x) = \begin{cases} 2x + 1, & x < 3 \\ 12 - x, & x \geq 3 \end{cases}$$

Find $f(3)$, and put it on the answer line.

Answer: _____

2. Using the same function f :

$$f(2) = \underline{\hspace{2cm}} \qquad f(5) = \underline{\hspace{2cm}}$$

3. Leo is asked for $g(4)$, where

$$g(x) = \begin{cases} x^2 - 2, & x \leq 4 \\ 3x + 1, & x > 4 \end{cases}$$

He writes $g(4) = 3(4) + 1 = 13$.

- (a) Name Leo's error. Say which condition decides the matter.
- (b) Find $g(4)$ correctly, and put it on the answer line.

Answer: _____

4. Still using g from Problem 3, suppose $g(x) = 23$ and the first rule is the one being tried.

- (a) Solve $x^2 - 2 = 23$ and list both values.
- (b) Which of your two values is a genuine solution of $g(x) = 23$? Explain how the condition $x \leq 4$ decides it.

Answer: _____

5. A piecewise function is defined by

$$h(x) = \begin{cases} 3x + k, & x < 2 \\ x^2 + 1, & x \geq 2 \end{cases}$$

Find the value of k that makes the two rules agree at the boundary, so the graph has no jump at $x = 2$.

Answer: _____

6. A car park posts the price list

$$C(t) = \begin{cases} 2t + 2, & 0 \leq t \leq 2 \\ 4t, & t > 2 \end{cases}$$

where t is the stay in hours and C is in dollars. Sofia copies the second condition as $t \geq 2$ instead of $t > 2$, and then cannot decide whether a two-hour stay costs \$6 or \$8.

- (a) Name the error in Sofia's version of the conditions, and explain why a price list written her way could not be a function.
- (b) Using the list exactly as printed above, find the cost of a two-hour stay.
- (c) Find the cost of a three-hour stay.

Answer: _____

7. A piecewise function is defined by

$$p(x) = \begin{cases} cx, & x < 5 \\ 2x + 5, & x \geq 5 \end{cases}$$

Find the value of c for which the graph of p has no jump at $x = 5$.

Answer: _____

8. **Synthesis challenge.** A piecewise function is defined by

$$q(x) = \begin{cases} x + a, & x \leq -1 \\ x^2, & x > -1 \end{cases}$$

- (a) Find the value of a for which the two rules agree at $x = -1$.
- (b) Using that value of a , find $q(-1)$ and $q(2)$.
- (c) Explain which rule supplies $q(-1)$ and why, then say what would go wrong if both conditions were written with $\leq$.

Answer: _____